Project Initialization and Planning Phase

Date	06 July 2026
Team ID	TMID-SL-2026-01
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine-learning-based credit model and real-time decision-making.

Project Overview

Field	Details
Objective	Revolutionize the loan approval process using machine learning, ensuring faster and more accurate eligibility asses
Scope	Comprehensively assess and enhance the loan approval process — from data collection and preprocessing throug

Problem Statement

Field	Details
Description	Inaccuracies and inefficiencies in the current manual loan-approval process adversely affect operational efficiency
Impact	Solving this improves operational efficiency, reduces lending risk, and improves customer satisfaction and organiza

Proposed Solution

Field	Details
Approach	Use machine learning (Decision Tree, Random Forest, KNN, XGBoost) to analyze and predict applicant creditworth
Key Features	• ML-based credit assessment model (XGBoost, selected after comparison) • Real-time decision-making for quicker loan approvals • SMOTE-balanced training data to avoid biased predictions • Retrainable pipeline that adapts as new data becomes available

Resource Requirements

Resource Type	Description	Specification / Allocation
Computing Resources	CPU specifications, number of cores	Standard multi-core CPU (no GPU required for these model sizes
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	< 1 GB (dataset + model artifacts)
Frameworks	Python web framework	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn, imbalanced-lear
Development Environment	IDE / notebook	Jupyter Notebook, PyCharm, Anaconda Navigator
Data	Source, size, format	Loan eligibility dataset, 614 rows, CSV